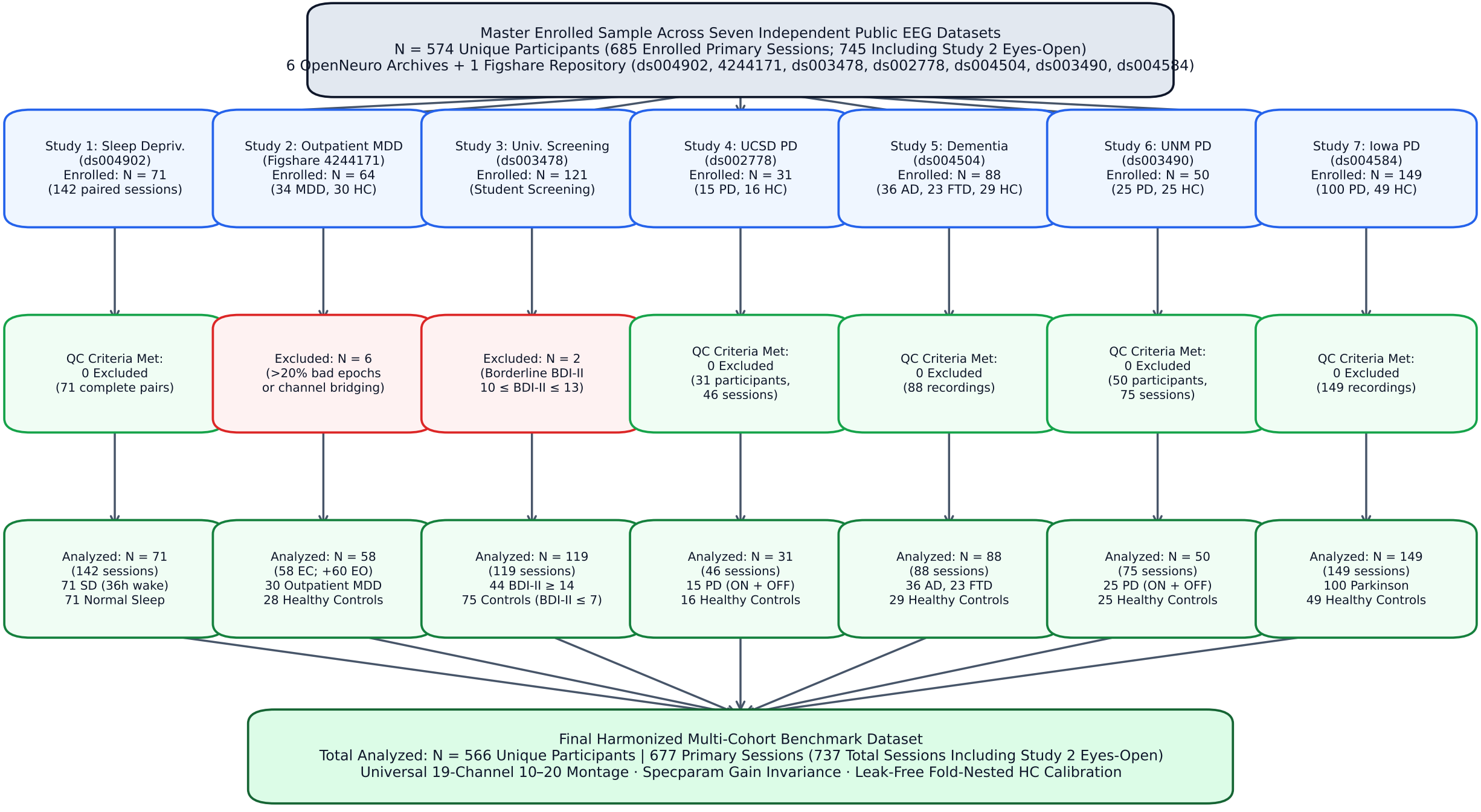
Master Enrolled Sample Across Seven Independent Public EEG Datasets
N = 574 Unique Participants (685 Enrolled Primary Sessions; 745 Including Study 2 Eyes-Open)
6 OpenNeuro Archives + 1 Figshare Repository (ds004902, 4244171, ds003478, ds002778, ds004504, ds003490, ds004584)
Study 1: Sleep Depriv. (ds004902) Enrolled: N = 71 (142 paired sessions)
Study 2: Outpatient MDD (Figshare 4244171) Enrolled: N = 64 (34 MDD, 30 HC)
Study 3: Univ. Screening (ds003478) Enrolled: N = 121 (Student Screening)
Study 4: UCSD PD (ds002778) Enrolled: N = 31 (15 PD, 16 HC)
Study 5: Dementia (ds004504) Enrolled: N = 88 (36 AD, 23 FTD, 29 HC)
Study 6: UNM PD (ds003490) Enrolled: N = 50 (25 PD, 25 HC)
Study 7: Iowa PD (ds004584) Enrolled: N = 149 (100 PD, 49 HC)
QC Criteria Met: 0 Excluded (71 complete pairs)
Excluded: N = 6 (>20% bad epochs or channel bridging)
Excluded: N = 2 (Borderline BDI-II 10 ≤ BDI-II ≤ 13)
QC Criteria Met: 0 Excluded (31 participants, 46 sessions)
QC Criteria Met: 0 Excluded (88 recordings)
QC Criteria Met: 0 Excluded (50 participants, 75 sessions)
QC Criteria Met: 0 Excluded (149 recordings)
Analyzed: N = 71 (142 sessions) 71 SD (36h wake) 71 Normal Sleep
Analyzed: N = 58 (58 EC; +60 EO) 30 Outpatient MDD 28 Healthy Controls
Analyzed: N = 119 (119 sessions) 44 BDI-II ≥ 14 75 Controls (BDI-II ≤ 7)
Analyzed: N = 31 (46 sessions) 15 PD (ON + OFF) 16 Healthy Controls
Analyzed: N = 88 (88 sessions) 36 AD, 23 FTD 29 Healthy Controls
Analyzed: N = 50 (75 sessions) 25 PD (ON + OFF) 25 Healthy Controls
Analyzed: N = 149 (149 sessions) 100 Parkinson 49 Healthy Controls
Final Harmonized Multi-Cohort Benchmark Dataset
Total Analyzed: N = 566 Unique Participants | 677 Primary Sessions (737 Total Sessions Including Study 2 Eyes-Open)
Universal 19-Channel 10–20 Montage · Specparam Gain Invariance · Leak-Free Fold-Nested HC Calibration